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REPLACEABLE LIP COSMETIC CONTAINER AND REPLACEMENT DEVICE

Abstract

A replaceable lip cosmetic container includes a top cover, a tube body, a guide member and a rotating seat, and further includes a push-out member; an inner wall of the tube body is provided with a spiral groove; the guide member is provided with a guide groove in an axial direction; an outer wall of the push-out member is provided with a spiral rod, the push-out member is sleeved in the guide member, and the spiral rod penetrates through the guide groove and cooperates with the spiral groove to form a spiral pair; the push-out member moves along the guide groove to be located at a storage position, a use position and a replacement position; and the guide groove is provided with a storage groove, a limit groove and a stop groove corresponding to the above positions, respectively.

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Background/Summary

CROSS-REFERENCE TO THE RELATED APPLICATIONS [0001] This application is the national phase entry of International Application No. PCT/CN2023/084563, filed on Mar. 29, 2023, which is based upon and claims priority to Chinese Patent Application No. 202211577389.7, filed on Dec. 9, 2022, the entire contents of which are incorporated herein by reference.

TECHNICAL FIELD
[0002] The present invention relates to the field of cosmetics, and particularly to a cosmetic container and a replacement device.
BACKGROUND
[0003] Most lipsticks are disposable. Specifically, the packaging materials with the remaining lipstick core are usually completely discarded after use and cannot be reused; this will cause extreme waste of packaging materials. In addition, the packaging materials are usually non-degradable, and a large amount of waste packaging materials seriously pollute the environment.
[0004] In order to solve this problem, the Chinese patent (CN113367465A) discloses a container structure capable of replacing lipstick paste. In this structure, the combining member is detachably connected to the lipstick assembly. The lipstick assembly can be completely replaced and recycled after use, and the base, combining member and upper cover can be reused repeatedly. Additionally, since the lipstick assembly is made of polyethylene terephthalate (PET), which is an environmentally friendly and recyclable material, it has the effect of reducing environmental pollution.
[0005] Existing replaceable lipsticks need to replace the lipstick assembly as a whole, and the replaced assembly contains structural parts that fit in motion, so it is necessary to use materials with high hardness, rigidity and durability such as PET. Therefore, they can only be recycled. However, when the recycling system is not fully covered and efficient, the lipstick assembly that is directly discarded will also cause environmental pollution.
SUMMARY
Technical Problems
[0006] The technical problem to be solved by the present invention is to provide a replaceable lip cosmetic container and a replacement device. In the cosmetic container, by the cooperation between a guide member and a push-out member, the push-out member has a replacement position that is convenient to assemble and disassemble the replacement device, thereby realizing repeated recycling of the cosmetic container. The replacement device only replaces the paste and necessary connection seat, and the replacement position can provide the reverse locking force required for installation. The operation is simple and the waste of materials is small. The replacement device is made of a degradable material, which will not pollute the environment even if it is directly discarded without being recycled, further meeting environmental protection requirements.
Technical Solutions
[0007] The present invention is realized as follows: A replaceable lip cosmetic container includes a top cover, a tube body, a guide member and a rotating seat. Two ends of the tube body are open; the top cover is covered at a top opening of the tube body; the rotating seat is hinged to a bottom opening of the tube body and forms a rotating pair with the tube body; the guide member is sleeved in the tube body. The replaceable lip cosmetic container further includes a push-out member; an inner wall of the tube body is provided with a spiral groove; the guide member is provided with a

guide groove in an axial direction; an outer wall of the push-out member is provided with a spiral rod, the push-out member is sleeved in the guide member, and the spiral rod penetrates through the guide groove and cooperates with the spiral groove to form a spiral pair; the push-out member moves along the guide groove to be located at a storage position, a use position and a replacement position; the guide groove is provided with a storage groove, a limit groove and a stop groove corresponding to the storage position, the use position and the replacement position, respectively; and a thread lead angle of the spiral groove is greater than or equal to an equivalent friction angle of the spiral pair.

[0008] The push-out member is a cylindrical cavity with one side open; and a connection boss is arranged at a center of a bottom of the cavity of the push-out member, and an outer wall of the connection boss is provided with a rib.

[0009] The connection boss is cylindrical, and the rib is trapezoidal in cross section.

[0010] There is a pair of guide grooves, which are centrally symmetrically arranged on the guide member; and there are two spiral rods, each of which is matched with one guide groove.

[0011] The storage groove and the stop groove are elongated grooves extending radially along the guide member.

[0012] A direction in which the push-out member moves from the storage position to the replacement position is taken as a positive direction, the limit groove includes an entry section and an exit section, and an inclination angle of the entry section is smaller than an equivalent friction angle between the spiral rod and the entry section.

[0013] The limit groove is shaped as a circular arc or a triangle.

[0014] The limit groove is an elongated groove extending radially along the guide member.

[0015] An outer sleeve is further sleeved outside the tube body, and the outer sleeve is fixedly connected to the tube body.

[0016] A replacement device is provided. The replacement device cooperates with the replaceable lip cosmetic container, and the replacement device is made of a degradable material.

Beneficial Effects

[0017] In the replaceable lip cosmetic container provided by the present invention, by the cooperation between the guide member and the push-out member, the push-out member has a replacement position that is convenient to assemble and disassemble the replacement device, thereby realizing repeated recycling of the cosmetic container. The replacement device only replaces the paste and necessary connection seat, and the replacement position can provide the reverse locking force required for installation. The operation is simple and the waste of materials is small. Additionally, in the use position, the guide groove is further provided with corresponding limit groove to facilitate normal use by users, so that the present invention not only meets environmental protection requirements, but also has wide promotion and application value.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0018] FIG. **1** is an exploded view showing components of a replaceable lip cosmetic container of the present invention;

[0019] FIG. **2** is a schematic diagram showing a structure of a tube body in the present invention;

[0020] FIG. **3** is a schematic diagram showing a structure of a guide member in the present invention;

[0021] FIG. **4** is a schematic diagram showing a structure of a rotating seat in the present invention;

[0022] FIG. **5** is a schematic diagram showing a structure of a push-out member in the present invention;

[0023] FIG. **6** is a schematic diagram showing a structure of a limit groove in an embodiment of

the present invention;
[0024] FIG. **7** is a schematic diagram showing a structure of a limit groove in another embodiment of the present invention;
[0025] FIG. **8** is a schematic diagram showing a structure of a replacement device in the first embodiment of the present invention;
[0026] FIG. **9** is a schematic diagram showing a structure of a replacement device in the second embodiment of the present invention;
[0027] FIG. **10** is a schematic diagram showing a structure of a replacement device in the third embodiment of the present invention; and
[0028] FIG. **11** is a schematic diagram showing a structure of a replacement device in the fourth embodiment of the present invention.
[0029] In the figures: **1** top cover, **2** tube body, **3** guide member, **4** rotating seat, **5** push-out member, **6** replacement device, **21** spiral groove, **22** outer sleeve, **31** guide groove, **32** storage groove, **33** limit groove, **34** stop groove, **51** spiral rod, **52** connection boss, **53** rib, **61** housing, **62** connection seat, **63** paste, **64** connection hole, **331** entry section, **332** exit section, **611** upper housing, **612** lower housing, **613** support hole, **614** base, **615** sealing cover, **616** tearing portion, **617** cutting line, **621** foundation cavity, and **622** transition cavity.

DETAILED DESCRIPTION OF THE EMBODIMENTS

[0030] The present invention will be further explained below in conjunction with specific embodiments. It should be understood that these embodiments are only used to illustrate the present invention without limiting the scope of the present invention. In addition, it should be understood that after reading the content of the present invention, those skilled in the art may make various changes or modifications to the present invention, and these equivalent forms also fall within the scope that is defined in the claims appended to this application.

Embodiments

[0031] As shown in FIGS. **1**, **2**, **3** and **4**, a replaceable lip cosmetic container includes the top cover **1**, the tube body **2**, the guide member **3** and the rotating seat **4**. Two ends of the tube body **2** are open. The top cover **1** is covered at the top opening of the tube body **2**. The rotating seat **4** is hinged to the bottom opening of the tube body **2** and forms a rotating pair with the tube body **2**. The guide member **3** is sleeved in the tube body **2**. The replaceable lip cosmetic container further includes the push-out member **5**. The inner wall of the tube body **2** is provided with the spiral groove **21**. The guide member **3** is provided with the guide groove **31** in the axial direction. The outer wall of the push-out member **5** is provided with the spiral rod **51**, the push-out member **5** is sleeved in the guide member **3**, and the spiral rod **51** penetrates through the guide groove **31** and cooperates with the spiral groove **21** to form a spiral pair. The push-out member **5** moves along the guide groove **31** to be located at a storage position, a use position and a replacement position. The guide groove **31** is provided with the storage groove **32**, the limit groove **33** and the stop groove **34** corresponding to the storage position, the use position and the replacement position, respectively.
[0032] In the present invention, in order to ensure operating comfort during use and allow the user to easily screw out the cosmetics installed on the push-out member **5**, the thread lead angle of the spiral groove **21** is greater than or equal to the equivalent friction angle of the spiral pair. At this time, the spiral pair does not have a self-locking function. Therefore, it is necessary to provide the limit groove **33** at the use position of the cosmetics to limit the retreat of the push-out member **5** at this position.
[0033] As shown in FIG. **8**, a replacement device for a replaceable lip cosmetic container is provided. The replacement device cooperates with the replaceable lip cosmetic container and includes the housing **61**, the connection seat **62** and the paste **63**. The housing **61** covers the paste **63**. The connection seat **62** is provided with the foundation cavity **621**. The bottom of the paste **63** is fixedly connected in the foundation cavity **621**. The use section of the paste **63** extends out of the connection seat **62**. The connection hole **64** is formed at the bottom of the connection seat **62**, and

the connection hole **64** is communicated with the foundation cavity **621** through the transition cavity **622**. The transition cavity **622** is allowed to provide an installation space that cooperates with the push-out member **5** and reduce the volume of the unusable part of the paste **63**, so as to reduce waste of the paste **63**.
[0034] In this embodiment, preferably, the base **614** is further provided outside the lower housing **612**, and the lower housing **612** is provided inside the base **614** to facilitate transportation and placement of the replacement device.
[0035] In the present invention, the replacement device is made of a degradable material; The degradable material is, for example, prepared by papermaking using bamboo fiber, wood fiber and other degradable fiber, and is required to meet the disposable needs; the part after use and replacement can be directly discarded and allowed to degrade naturally. It will not cause pollution to the environment like conventional plastics, and can meet environmental protection requirements even if it is not recycled.
[0036] Further, in the present invention, the foundation cavity **621** and the transition cavity **622** are cylindrical cavities, the connection hole **64** is a round hole, and the foundation cavity **621**, the transition cavity **622** and the connection hole **64** are coaxially arranged.
[0037] As shown in FIG. **9**, in this embodiment, the diameter of the cross-section circle of the transition cavity **622** is smaller than the diameter of the cross-section circle of the foundation cavity **621**; the smaller transition cavity **622** is retracted inwards to form a step to support the paste **63**; the diameter of the connection hole **64** is smaller than the diameter of the cross-section circle of the transition cavity **622**.
[0038] The present invention can be further described as that the housing **61** includes the upper housing **611** and the lower housing **612**. The lower housing **612** is provided with the support hole **613** that matches the outer shape of the connection seat **62**. The upper housing **611** covers the top of the lower housing **612**. The connection seat **62** is detachably installed in the support hole **613** to facilitate the cooperation and stress requirements between the support hole **613** and the connection seat **62**.
[0039] In the Embodiments 3 and 4 of the replacement device of the present invention, the present invention can be further described as that the housing **61** includes the upper housing **611** and the sealing cover **615**. The upper housing **611** covers the top of the connection seat **62** and covers the paste **63**, and the sealing cover **615** covers the connection hole **64**.
[0040] As shown in FIG. **10**, the sealing cover **615** is a protective film, and the edge of the protective film is provided with the tearing portion **616**.
[0041] As shown in FIG. **11**, the sealing cover **615** is a sealing cover sheet arranged on the connection hole **64** and integrally formed with the connection seat **62**. The edge of the sealing cover sheet is the cutting line **617**, and the sealing cover sheet is separated from the connection seat **62** along the cutting line **617** under the action of external force.
[0042] As shown in FIG. **5**, in the present invention, the replaceable lip cosmetic container cooperates with the replacement device **6**. In this embodiment, in order to facilitate the detachable connection of the replacement device **6** and the replaceable lip cosmetic container, the push-out member **5** is a cylindrical cavity with one side open. The connection boss **52** is arranged at the center of the bottom of the cavity of the push-out member **5**, and the outer wall of the connection boss **52** is provided with the rib **53**; the connection boss **52** and the connection hole **64** are in interference fit, so that the connection seat **62** is detachably installed in the cavity.
[0043] When the replacement device **6** is replaced, firstly, the rotating seat **4** is rotated to lift the push-out member **5** to the replacement position; next the housing **61** is opened to take out the connection seat **62** and the paste **63**; and then the connection seat **62** is installed on the push-out member **5** to achieve detachable installation. During installation, the connection seat **62** is pressed down to be tightly connected to the push-out member **5**; due to the existence of the storage groove **32**, the pressing action will not cause the push-out member **5** to descend, so as to ensure that the

installation action can be completed smoothly.
[0044] In the present invention, preferably, the connection boss **52** is cylindrical, and the rib **53** is trapezoidal in cross section, so as to provide great interference fit pressure and ensure the installation stability of the connection seat **62**.
[0045] Taking into account the stress stability when the push-out member **5** is rotated and jacked, there is a pair of guide grooves **31**, which are centrally symmetrically arranged on the guide member **3**; and there are two spiral rods **51**, each of which is matched with one guide groove **31**.
[0046] In the present invention, the storage groove **32** and the stop groove **34** are elongated grooves extending radially along the guide member **3**; the storage groove **32** prevents cosmetics from naturally extending out due to inverted placement when the present device is not in use, and the stop groove **34** provides the stopping force required when the replacement device is installed.
[0047] In the present invention, the direction in which the push-out member **5** moves from the storage position to the replacement position is taken as the positive direction, the limit groove **33** includes the entry section **331** and the exit section **332**, and the inclination angle of the entry section **331** is smaller than the equivalent friction angle between the spiral rod **51** and the entry section **331**. When the spiral rod **51** enters the limit groove **33**, the entry section **331** is allowed to provide a locking force that prevents the spiral rod **51** from moving in the negative direction. At this time, when the rotating seat is not rotated, the push-out member **5** will not be pushed by the force in the negative direction force to meet the pressure requirements of the device during normal use.
[0048] In this embodiment, preferably, the limit groove **33** is shaped as a circular arc or a triangle as shown in FIG. **6**.
[0049] As shown in FIG. **7**, in another embodiment, the limit groove **33** is an elongated groove extending radially along the guide member **3**.
[0050] Further, considering the overall aesthetics and durability of the product, the outer sleeve **22** is further sleeved outside the tube body **2**, and the outer sleeve **22** is fixedly connected to the tube body **2**.

Claims

1. A replaceable lip cosmetic container, comprising a top cover, a tube body, a guide member and a rotating seat, wherein two ends of the tube body are open; the top cover is covered at a top opening of the tube body; the rotating seat is hinged to a bottom opening of the tube body and forms a rotating pair with the tube body; the guide member is sleeved in the tube body; wherein the replaceable lip cosmetic container further comprises a push-out member; an inner wall of the tube body is provided with a spiral groove; the guide member is provided with a guide groove in an axial direction; an outer wall of the push-out member is provided with a spiral rod, the push-out member is sleeved in the guide member, and the spiral rod penetrates through the guide groove and cooperates with the spiral groove to form a spiral pair; the push-out member moves along the guide groove to be located at a storage position, a use position and a replacement position; the guide groove is provided with a storage groove, a limit groove and a stop groove corresponding to the storage position, the use position and the replacement position, respectively; and a thread lead angle of the spiral groove is greater than or equal to an equivalent friction angle of the spiral pair.
2. The replaceable lip cosmetic container according to claim 1, wherein the push-out member is a cylindrical cavity with one side open; and a connection boss is arranged at a center of a bottom of the cylindrical cavity of the push-out member, and an outer wall of the connection boss is provided with a rib.
3. The replaceable lip cosmetic container according to claim 2, wherein the connection boss is cylindrical, and the rib is trapezoidal in cross section.
4. The replaceable lip cosmetic container according to claim 1, wherein there is a pair of guide

grooves, which wherein the pair of guide grooves are centrally symmetrically arranged on the guide member; and there are two spiral rods, wherein each of the two spiral rods is matched with one of the pair of guide grooves.

5. The replaceable lip cosmetic container according to claim 4, wherein the storage groove and the stop groove are elongated grooves extending radially along the guide member.

6. The replaceable lip cosmetic container according to claim 4, wherein a direction where the push-out member moves from the storage position to the replacement position is taken as a positive direction, the limit groove comprises an entry section and an exit section, and an inclination angle of the entry section is smaller than an equivalent friction angle between the spiral rod and the entry section.

7. The replaceable lip cosmetic container according to claim 6, wherein the limit groove is shaped as a circular arc or a triangle.

8. The replaceable lip cosmetic container according to claim 6, wherein the limit groove is an elongated groove extending radially along the guide member.

9. The replaceable lip cosmetic container according to claim 4, wherein an outer sleeve is further sleeved outside the tube body, and the outer sleeve is fixedly connected to the tube body.

10. A replacement device, wherein the replacement device cooperates with the replaceable lip cosmetic container according to claim 1, and the replacement device is made of a degradable material.

11. The replaceable lip cosmetic container according to claim 2, wherein there is a pair of guide grooves, wherein the pair of guide grooves are centrally symmetrically arranged on the guide member; and there are two spiral rods, wherein each of the two spiral rods is matched with one of the pair of guide grooves.

12. The replaceable lip cosmetic container according to claim 3, wherein there is a pair of guide grooves, wherein the pair of guide grooves are centrally symmetrically arranged on the guide member; and there are two spiral rods, wherein each of the two spiral rods is matched with one of the pair of guide grooves.

13. The replacement device according to claim 10, wherein in the replaceable lip cosmetic container, the push-out member is a cylindrical cavity with one side open; and a connection boss is arranged at a center of a bottom of the cylindrical cavity of the push-out member, and an outer wall of the connection boss is provided with a rib.

14. The replacement device according to claim 13, wherein in the replaceable lip cosmetic container, the connection boss is cylindrical, and the rib is trapezoidal in cross section.

15. The replacement device according to claim 10, wherein in the replaceable lip cosmetic container, there is a pair of guide grooves, wherein the pair of guide grooves are centrally symmetrically arranged on the guide member; and there are two spiral rods, wherein each of the two spiral rods is matched with one of the pair of guide grooves.

16. The replacement device according to claim 15, wherein in the replaceable lip cosmetic container, the storage groove and the stop groove are elongated grooves extending radially along the guide member.

17. The replacement device according to claim 15, wherein in the replaceable lip cosmetic container, a direction where the push-out member moves from the storage position to the replacement position is taken as a positive direction, the limit groove comprises an entry section and an exit section, and an inclination angle of the entry section is smaller than an equivalent friction angle between the spiral rod and the entry section.

18. The replacement device according to claim 17, wherein in the replaceable lip cosmetic container, the limit groove is shaped as a circular arc or a triangle.

19. The replacement device according to claim 17, wherein in the replaceable lip cosmetic container, the limit groove is an elongated groove extending radially along the guide member.

20. The replacement device according to claim 15, wherein in the replaceable lip cosmetic container, an outer sleeve is further sleeved outside the tube body, and the outer sleeve is fixedly connected to the tube body.